Selected Works
2020 - 2026



Mariana Silva

Architecture Portfolio

Copenhagen | 2026

01. DTU Building
Educational Building | Copenhagen | 2025

02. TerraVista
Modular Housing | Cape Verde | 2024

03. Housing in Cascais
Residential | Lisbon | 2020

04. Comparative Analysis of Bioclimatic Techniques
Master's Thesis | Lisbon and Copenhagen | 2026

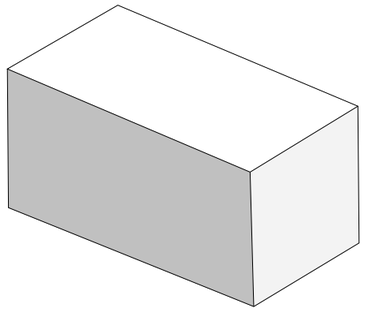


DTU Building

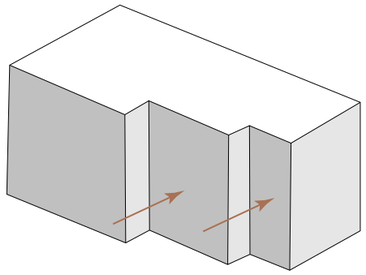
Educational Building | Copenhagen | 2025

Located at the Technical University of Denmark, this proposal explores the relationship between structure, façade and spatial organisation. A modular system defines flexible learning environments while maximising daylight, adaptability and construction efficiency.

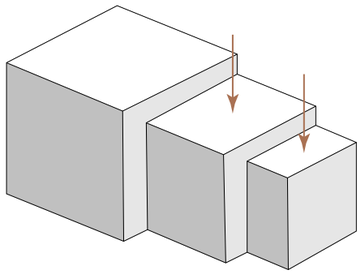
Concept Development



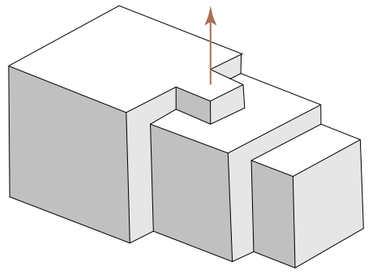
01. Initial Volume
Establishing the maximum buildable envelope defined by the site and urban context.



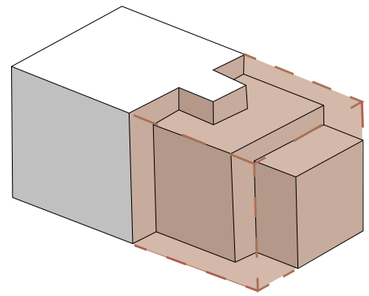
02. Light
Carving the mass to introduce daylight into the building.



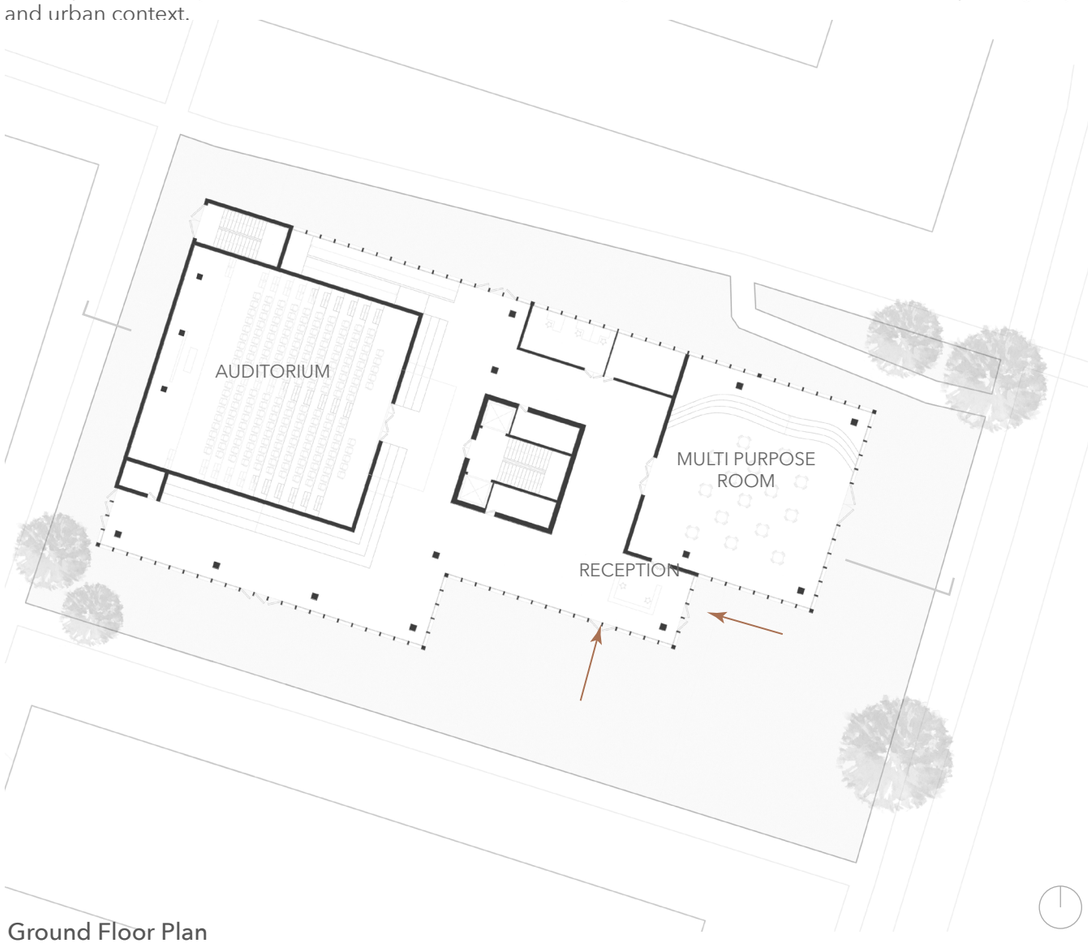
03. Open Roof
Creating accessible outdoor spaces while improving daylight to the surroundings.



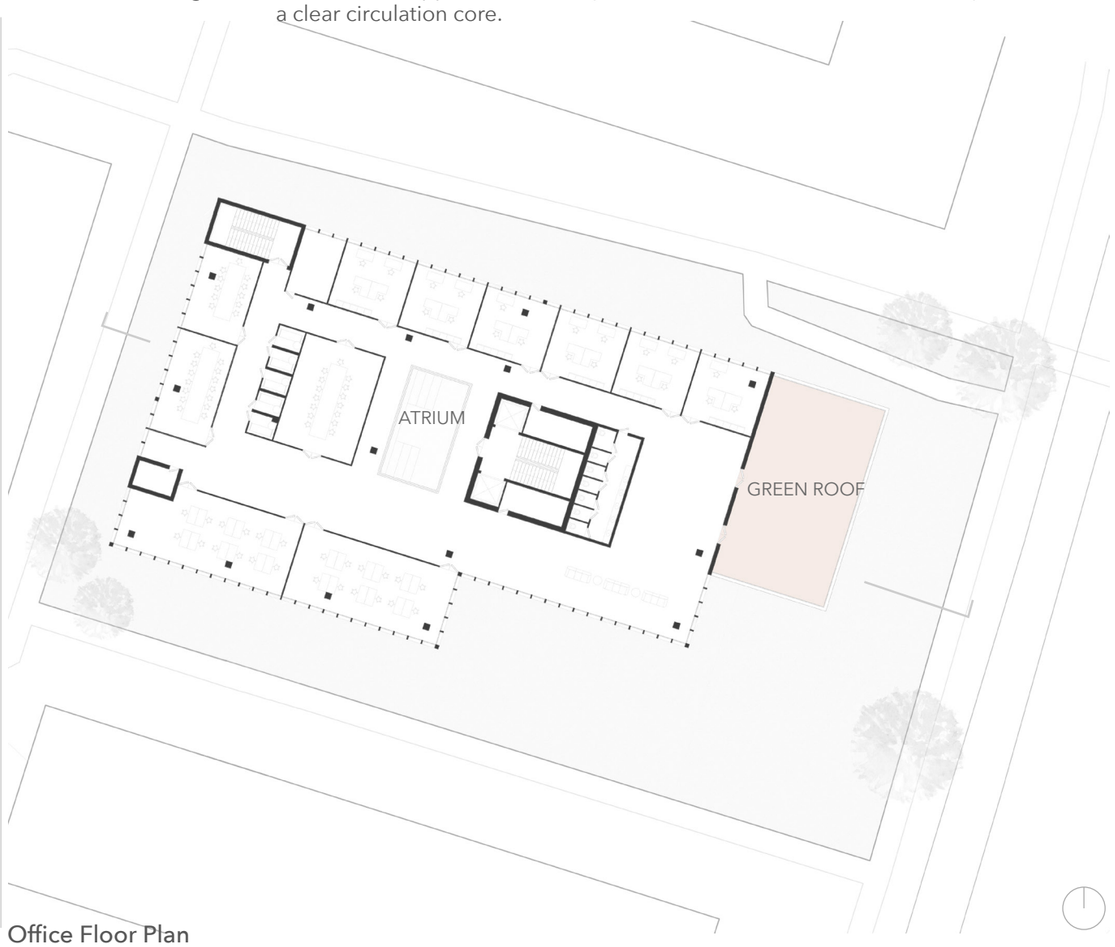
04. Access
Introducing vertical circulation, giving access to the upper levels through a clear circulation core.



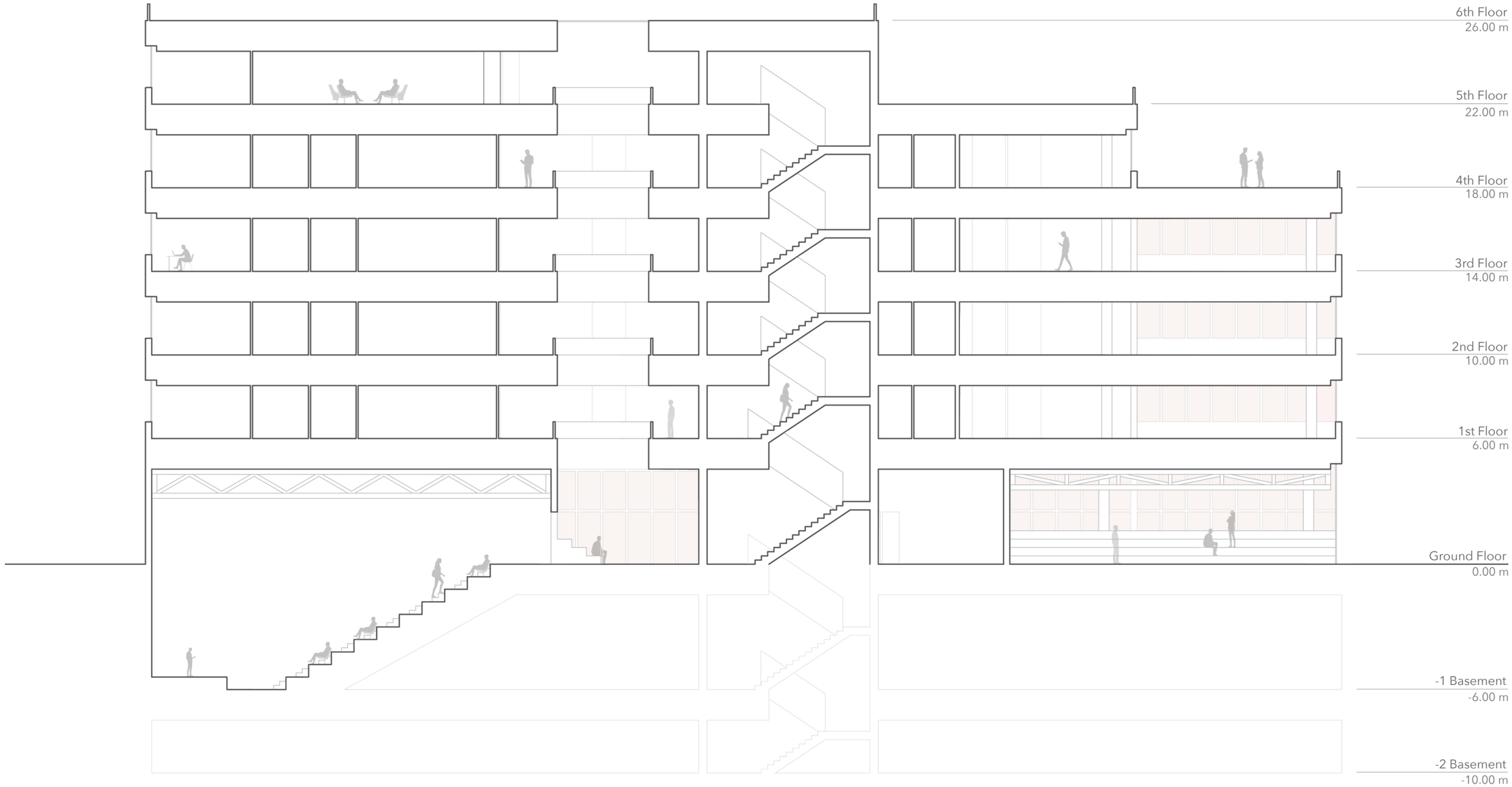
05. Future Scenario
Designing the structural grid to accommodate future growth.



Ground Floor Plan
Scale 1: 500



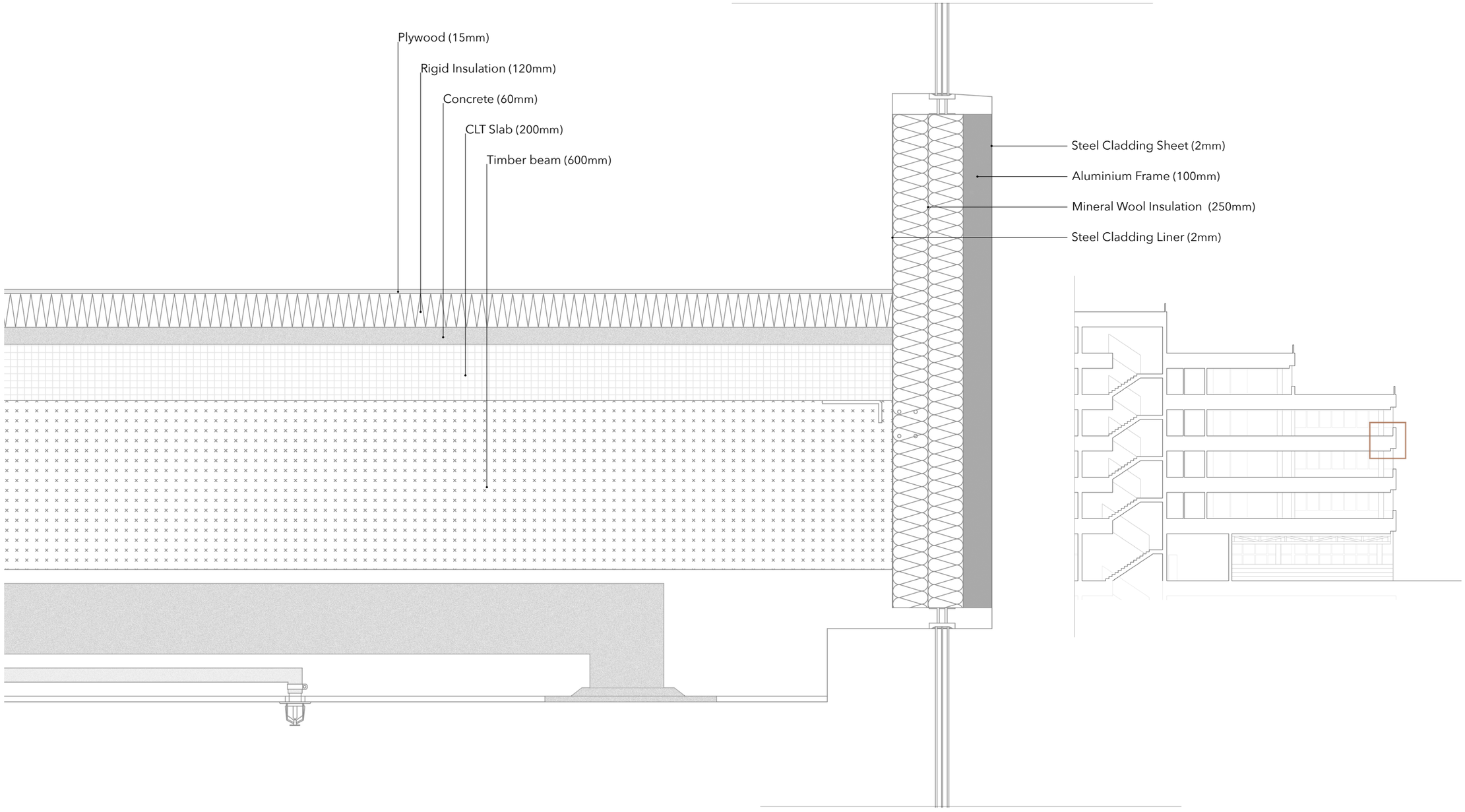
Office Floor Plan
Scale 1: 500



Longitudinal Section

Scale 1: 200

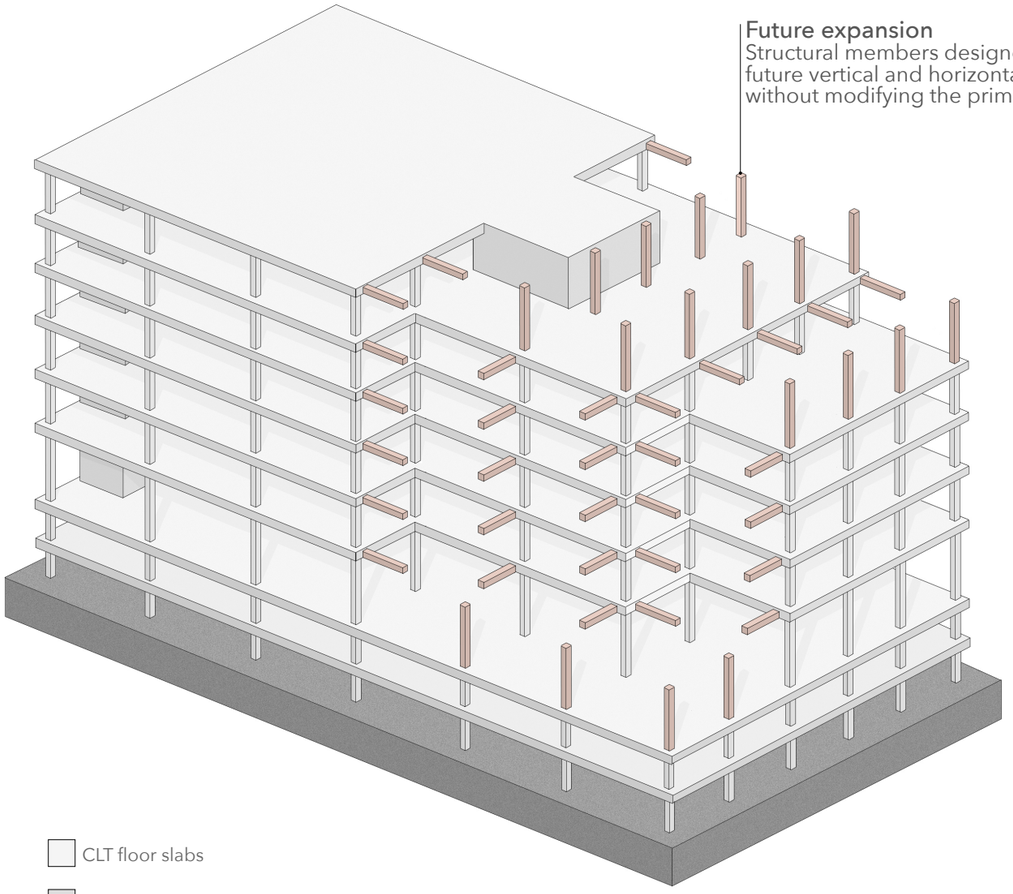
Construction Detail



Typical Façade-to-Floor Connection Detail
Scale 1:10

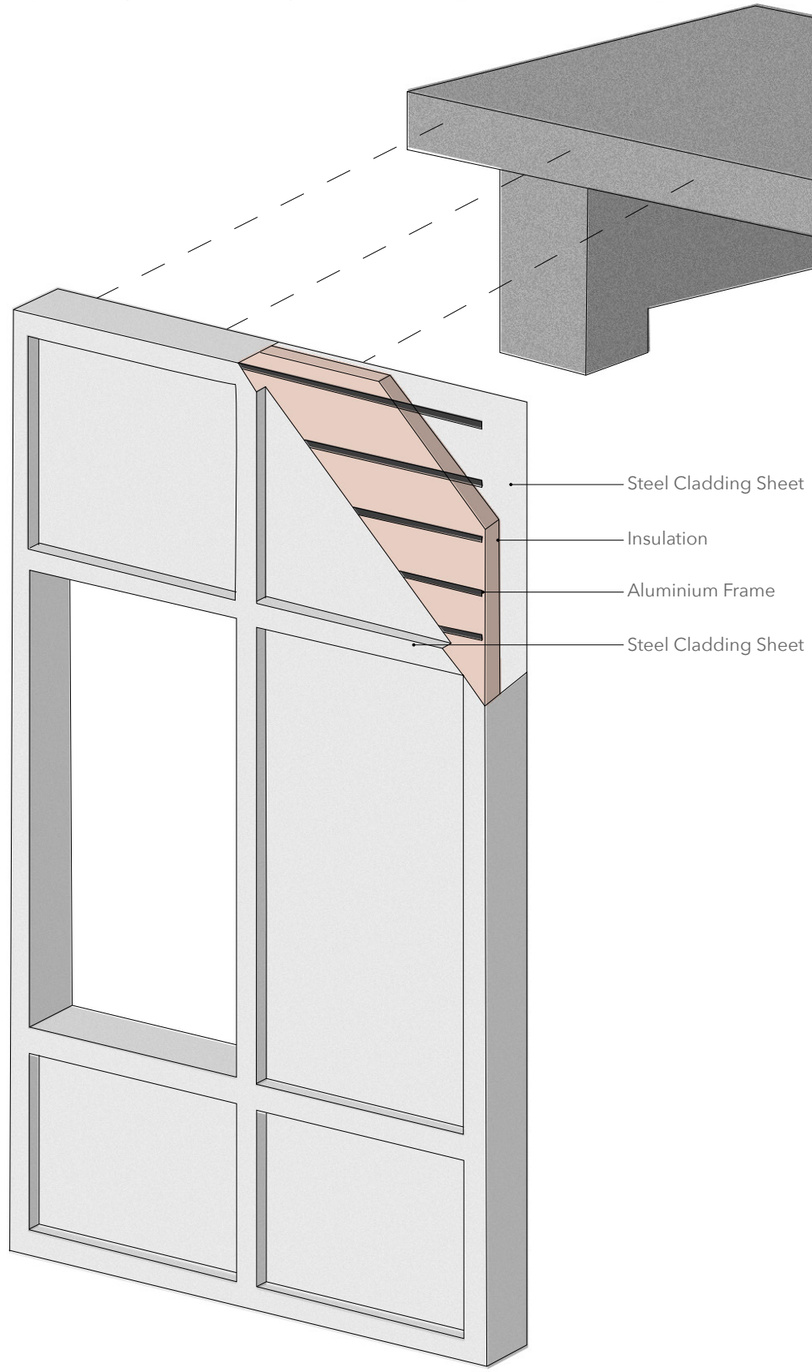
Structural System

The building is organized around a regular timber grid of CLT slabs, beams and columns, providing clear load paths and flexible open floor plates. The structural layout supports modular construction and enables future vertical expansion without altering the primary system.



Prefabricated Façade Module

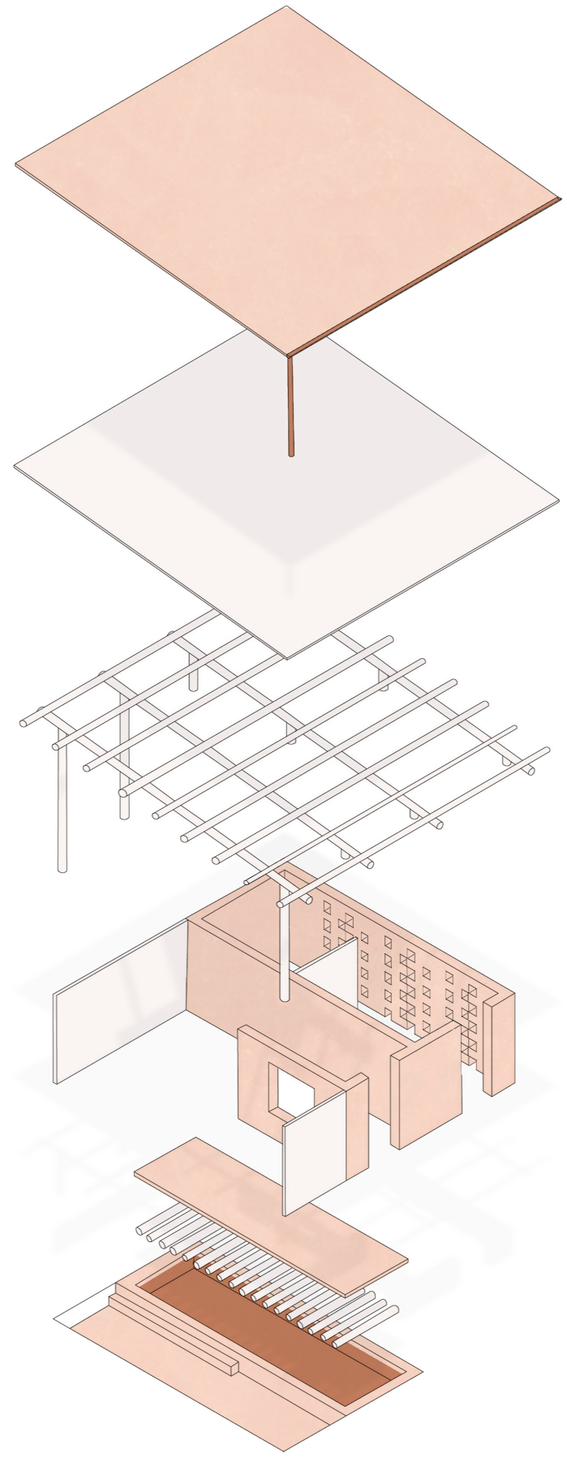
The façade consists of prefabricated steel panels fixed directly to the structural frame. The modular system reduces construction time, improves assembly precision and allows façade components to be replaced or reconfigured as the building evolves.



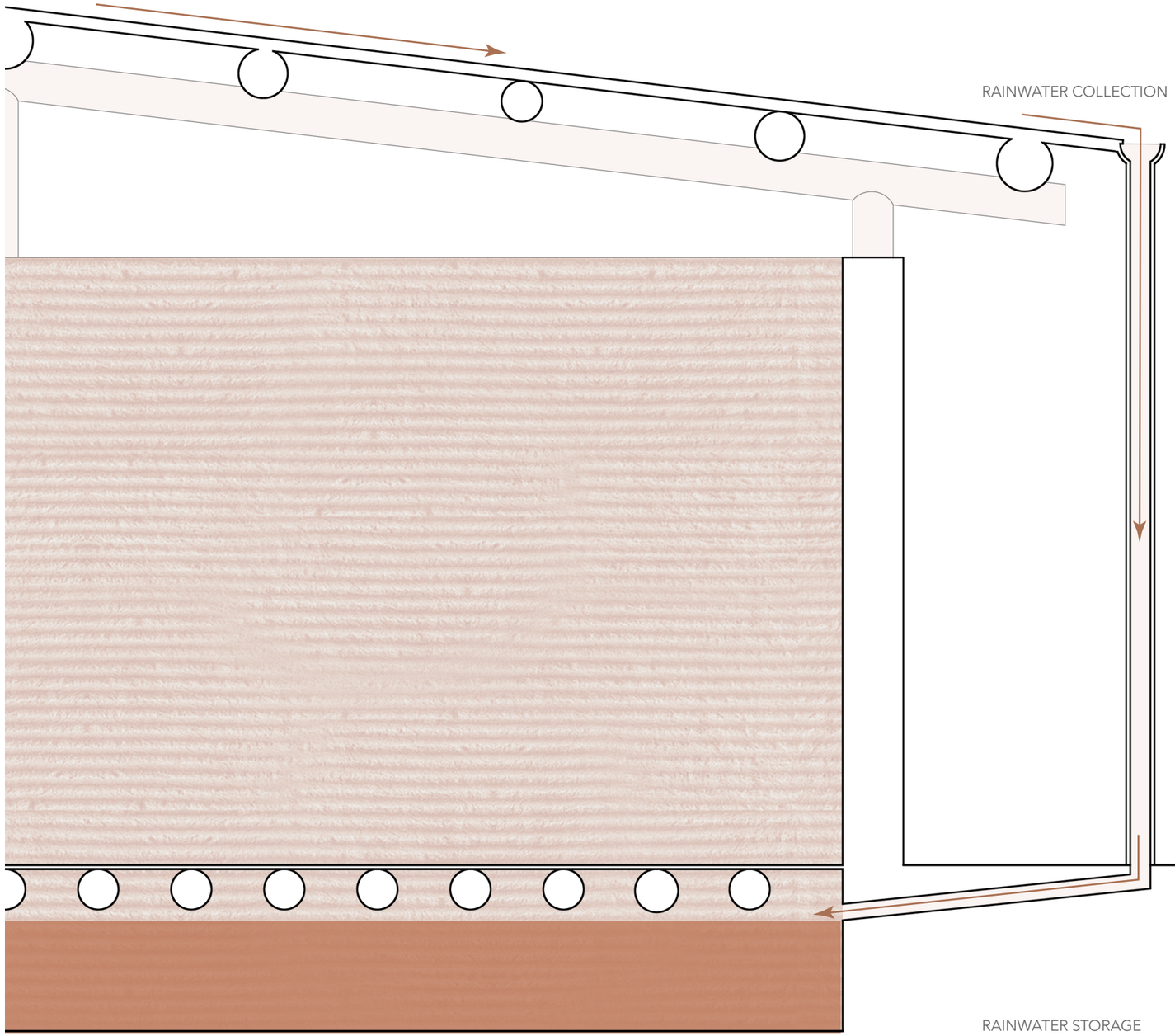
TerraVista

Modular Housing | Cape Verde | 2024

TerraVista is a modular housing proposal that combines rammed earth construction, locally sourced materials and passive environmental strategies. Designed to be easily built, expanded and maintained by local communities, it adapts to both the climate and the changing needs of its inhabitants.



Water and Material Strategy

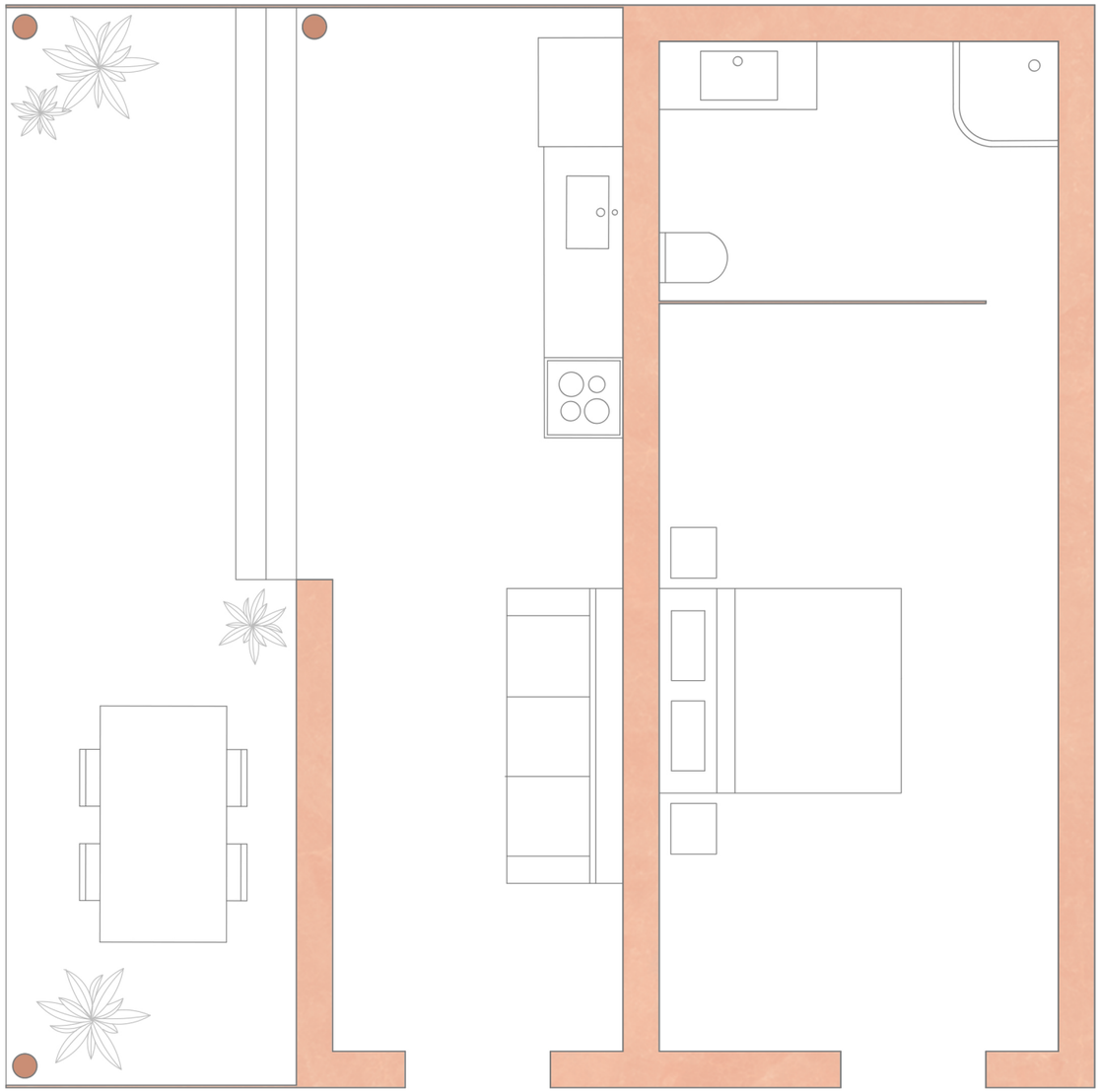


Water Collection Section
Scale 1: 25

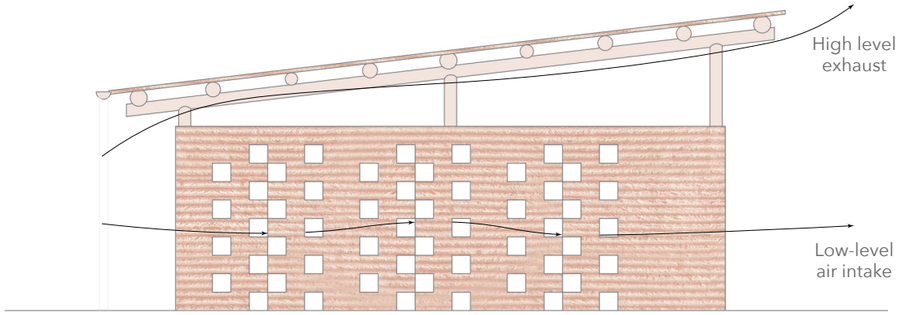
Local Material Palette



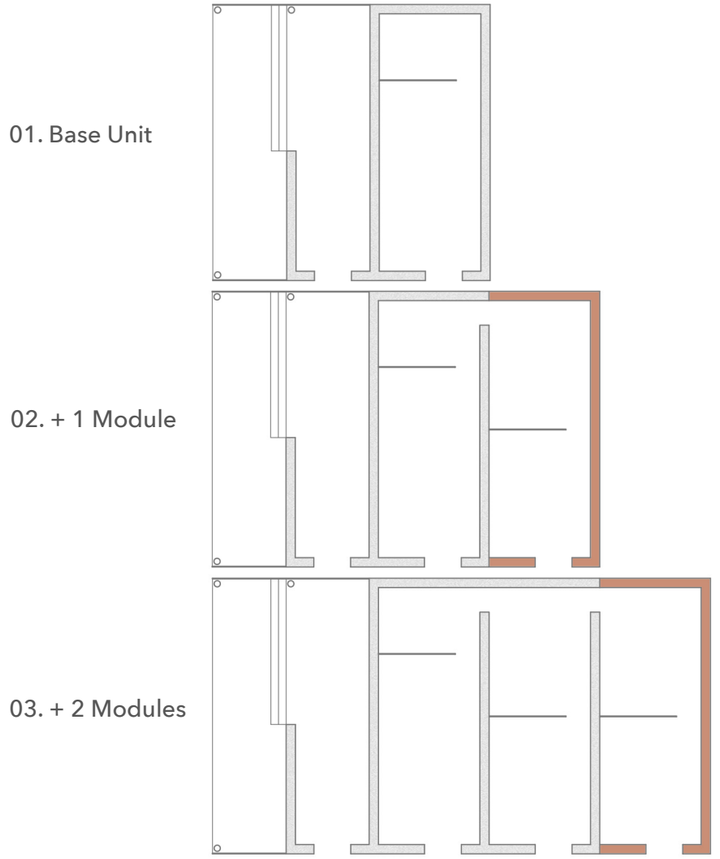
Environmental and Spatial Adaptability



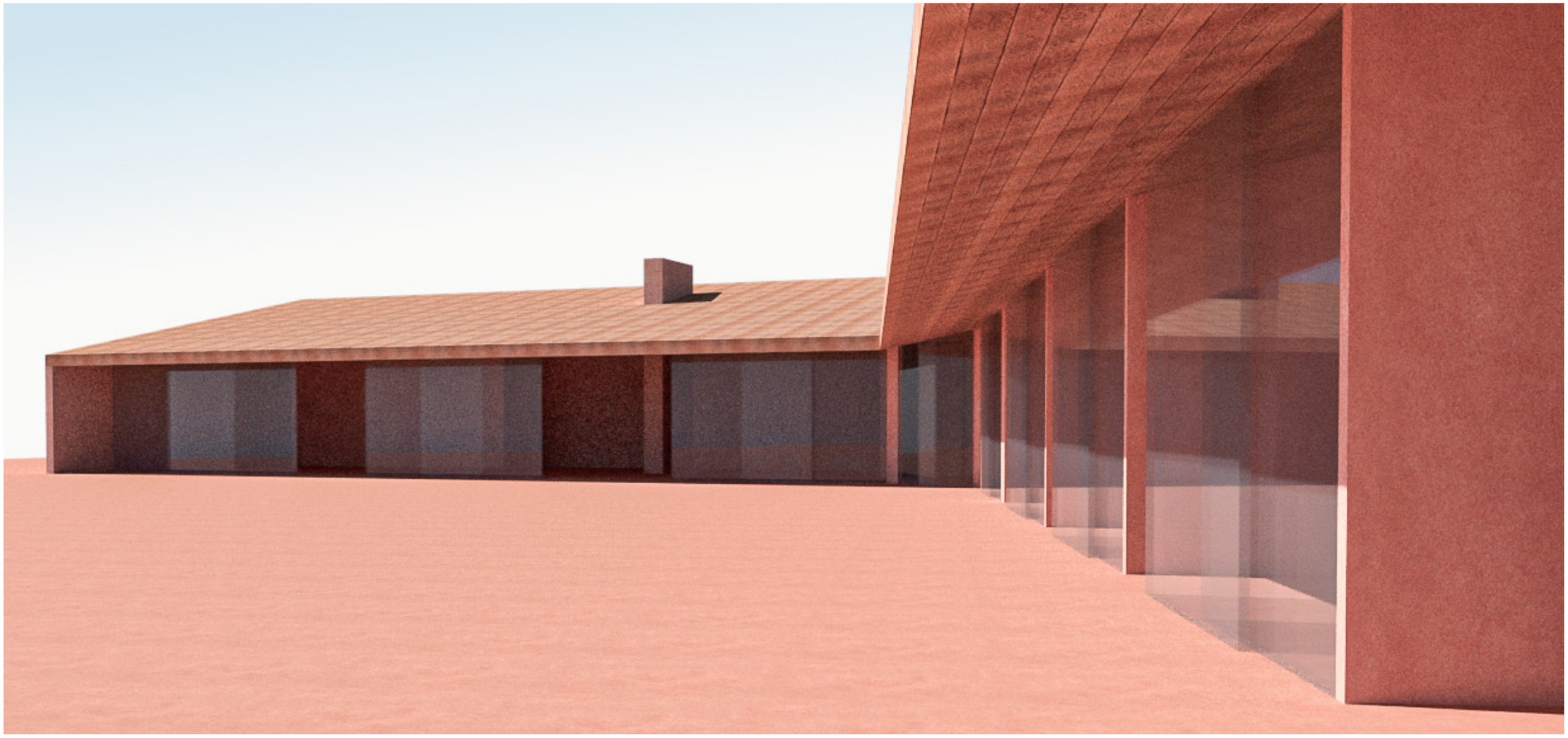
Floor Plan
Scale 1: 50



Natural Ventilation Section
Scale 1: 100



Incremental Modular Growth
Adaptable dwelling through the addition of spatial bays | Scale 1: 200



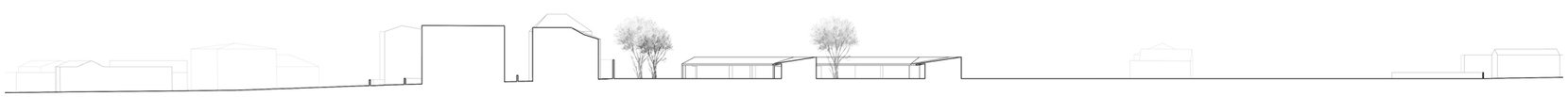
Housing in Cascais

Residential | Lisbon | 2020

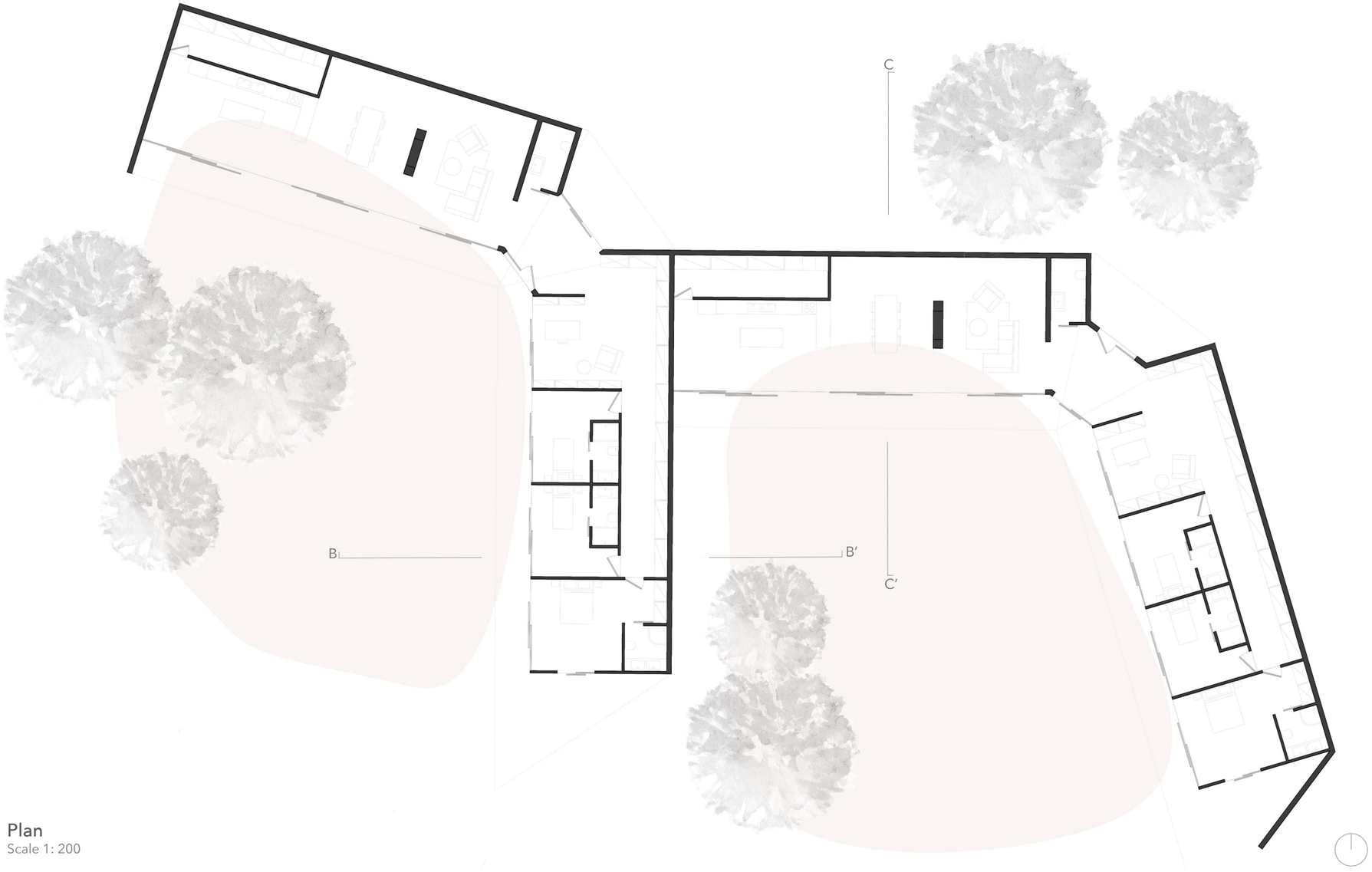
Located in Aldeia da Areia, Cascais, this project proposes a residential community where shared and private courtyards define the relationship between architecture and landscape. Each dwelling is organised around its own outdoor space, creating a sequence of light-filled interiors while ensuring privacy, spatial continuity and a strong connection to nature.



Site Plan
Scale 1:1000



Longitudinal Section A - A'
Scale 1:1000



Plan
Scale 1: 200



Section B - B′
Scale 1: 200

Natural Ventilation
Section C - C′ | Scale 1: 200

Comparative Analysis of Bioclimatic Techniques

Master's Thesis | Lisbon and Copenhagen | 2026

This thesis investigates how bioclimatic techniques adapt to different climatic conditions through a comparative analysis of 38 residential projects and two detailed case studies in Portugal and Denmark. The research examines how solar strategies, natural ventilation, envelope design and vegetation are combined to translate climatic conditions into architectural responses.

HOW ARE BIOCLIMATIC TECHNIQUES ADAPTED, COMBINED AND INTEGRATED ACROSS DIFFERENT CLIMATIC CONTEXTS?

Methodology

01. Literature Review
Study of bioclimatic principles, strategies and their application in architecture.

→

02. Climate Analysis
Global climate classification and detailed analysis of Lisbon and Aarhus.

→

03. Global Database
Analysis of 38 residential projects across 5 climatic contexts.

→

04. Case Studies
In-depth analysis of Torre Verde (Lisbon) and Sneglehusene (Aarhus).

→

05. Comparative Synthesis
Cross case comparison and identification of adaptive principles for the future.

Bioclimatic Strategies by Climate

Frequency of bioclimatic strategies across the 38 case studies

Climatic Context	Solar Gains	Solar Control	Natural Ventilation	Envelope	Vegetation
Cold Long, cold winters	81%	38%	33%	38%	47%
Temperate Mild seasons	83%	21%	71%	37%	37%
Mediterranean Hot, dry summers	71%	57%	90%	4%	23%
Hot-Arid Very hot, dry seasons	14%	90%	67%	28%	19%
Tropical Hot, humid all year	59%	67%	100%	7%	52%

Key Takeaways

01. Same principles, different weight
All climates use similar strategies, but their relative importance changes significantly.

02. Combination Matters
Strategies are combined and prioritised differently according to climatic context.

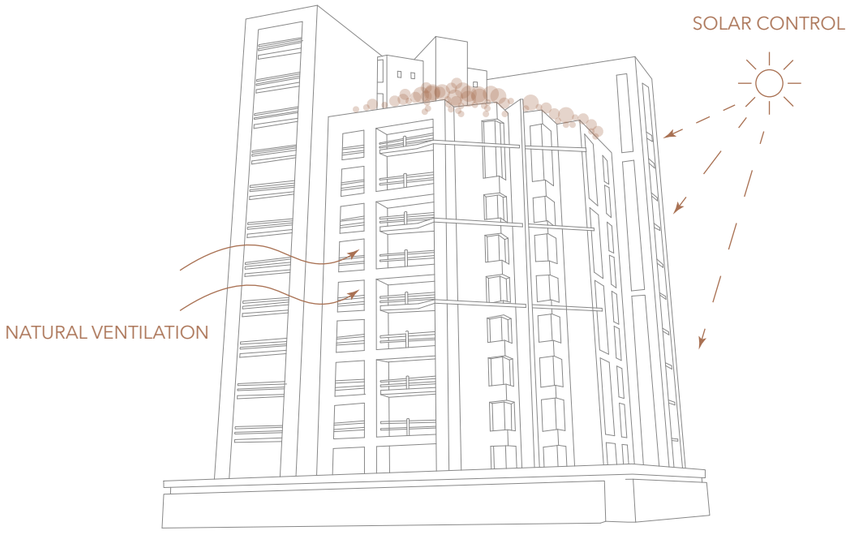
03. Context is essential
Climate drives architectural decisions in form, envelope, openings and landscape integration

Adaptation Across Climates

Two case studies in contrasting climates. Shared principles, different architectural responses.

Torre Verde

Lisbon, Portugal
Mediterranean Climate



Bioclimatic strategies respond to strong solar radiation and the need for heat dissipation. Solar protection and thermal mass are integrated with façade articulation and the spatial organization of individual apartments.

01. Solar Control
Solar protection and thermal mass regulate solar exposure and heat gains.

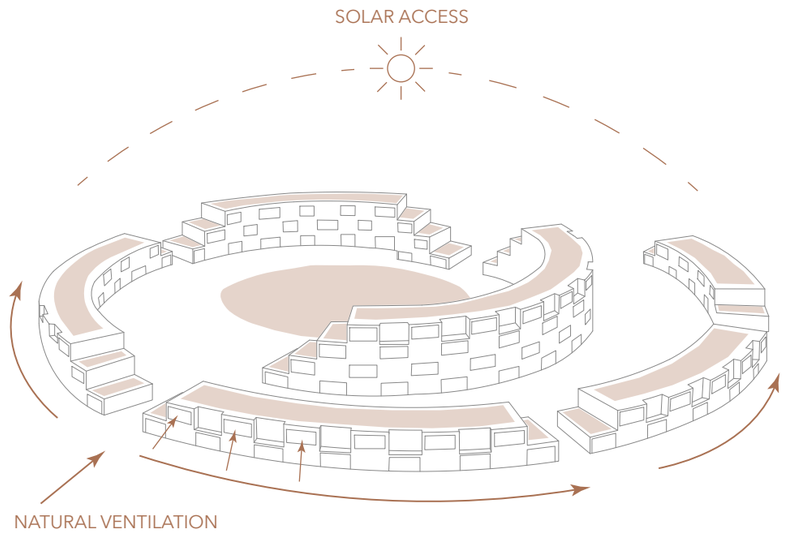
02. Natural Ventilation
Natural ventilation acts as a passive cooling strategy and influences the spatial organization of the apartments.

03. Envelope
Heavy construction, external insulation and integrated shading regulate heat gain and increase thermal inertia.

CONTROL + DISSIPATE

Sneglehusene

Aarhus, Denmark
Temperate Maritime Climate



The project responds to the need to minimise heat loss and maximise solar access. Building form, envelope design and ventilation work together to control environmental exposure.

01. Solar Gains
The stepped spiral configuration maximises solar exposure and daylight while preventing excessive self-shading.

02. Natural Ventilation
Cross ventilation provides air renewal, while the building form creates sheltered conditions within the spiral.

03. Envelope
Prefabricated concrete and an exterior pine layer create a continuous envelope that limits heat loss and protects against low temperatures and wind.

MAXIMISE + PROTECT